

03_Partial Derivative of Cost Function

1. Mean Squared Error (MSE)

$$J = \frac{1}{n} \sum (\hat{y} - y)^2$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{2}{n} (\hat{y} - y)$$

In the context of the MSE & MAE and most machine learning models:

- y is called the true value, actual value, or ground truth.
- $\hat{y}$ (pronounced "y-hat") is called the predicted value or model output.

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2. Mean Absolute Error (MAE)

$$J = \frac{1}{n} \sum |\hat{y} - y|$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{1}{n} \cdot \begin{cases} 1, & \hat{y} > y \\ -1, & \hat{y} < y \\ 0, & \hat{y} = y \end{cases}$$

Notes

The derivative of $|\hat{y} - y|$ depends on the sign of $\hat{y} - y$:

- If $\hat{y} > y$, then $|\hat{y} - y| = \hat{y} - y$ and the derivative w.r.t. $\hat{y}$ is 1.
- If $\hat{y} < y$, then $|\hat{y} - y| = -(\hat{y} - y) = y - \hat{y}$ and the derivative w.r.t. $\hat{y}$ is -1.
- If $\hat{y} = y$, the derivative is 0 (no error, hence no change needed).

3. Binary Cross-Entropy (Log Loss)

Used for **binary classification**, where:

- $y \in \{0,1\}$
- $\hat{y} \in (0,1)$ (usually from sigmoid)

Definition

$$J = -[y \log (\hat{y}) + (1 - y) \log (1 - \hat{y})]$$

Step 1: Differentiate term by term.

1. $\frac{d}{d\hat{y}} [-y \log (\hat{y})] = -y \cdot \frac{1}{\hat{y}} = -\frac{y}{\hat{y}}$
2. $\frac{d}{d\hat{y}} [-(1 - y) \log (1 - \hat{y})] = -(1 - y) \cdot \frac{-1}{1 - \hat{y}} = \frac{1 - y}{1 - \hat{y}}$

Step 2: Combine them:

$$\frac{dJ}{d\hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}}$$

Derivative (w.r.t. $\hat{y}$)

$$\frac{\partial J}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

Intuition

- Penalizes confident wrong predictions heavily
- Probabilistic interpretation
- Standard loss for logistic regression

4. Hinge Loss

Used in **SVMs** and margin-based classifiers.

Definition

$$J = \max(0, 1 - y\hat{y})$$

where:

- $y \in \{-1, +1\}$

Derivative (w.r.t. $\hat{y}$)

$$\frac{\partial J}{\partial \hat{y}} = \begin{cases} 0 & \text{if } y\hat{y} \geq 1 \\ -y & \text{otherwise} \end{cases}$$

Intuition

- Encourages a **margin**, not just correctness
- No penalty once prediction is “good enough”

5. Kullback–Leibler (KL) Divergence

KL divergence measures how one **probability distribution** differs from another.

Definition

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

Equivalent form:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) = \sum_x P(x) [\log P(x) - P(x) \log Q(x)]$$

Only the **second term** depends on $Q(x)$.

Where:

- $P(x)$ = true distribution
- $Q(x)$ = predicted distribution (e.g., softmax output)

Focus on:

$$-P(x) \log Q(x)$$

Derivative:

$$\frac{\partial}{\partial Q(x)} [-P(x) \log Q(x)] = -P(x) \cdot \frac{1}{Q(x)}$$

Final result

$$\boxed{\frac{\partial D_{KL}(P \parallel Q)}{\partial Q(x)} = -\frac{P(x)}{Q(x)}}$$